A probabilistic approach to the density of tautologies

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Propositional Logic System

Language	Logic
Syntax	Well-formed formulae
Semantics	Valuations
Syntactics	Proofs
Semantic Truth	Tautologies
Syntactic Entailment	Theorems

Well-formed formulae

- Variables

  x_0, x_1, $\cdots$,

  $\overline{x_0}$, $\overline{x_1}$, $\cdots$,

  $\bot$, $\top$

  x, $x1$, $x11$, $x111$, $\cdots$

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- Logical operators (Connectives)

  $\neg$

  $\rightarrow, \vee, \wedge, |, \cdots$

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- Logical operators (Connectives)

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  $\rightarrow$, $\vee$, $\wedge$, $|$, $\cdots$

Number of Well-formed formulae

If we have m variables, $\neg$, $\rightarrow$, there exists

$\neg\neg\neg\neg X$ type : m

$\neg\neg[X \rightarrow Y]$ type : m^2

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$[\neg X \rightarrow \neg Y]$ type : m^2

$[\neg\neg X \rightarrow Y]$ type : m^2

$[X \rightarrow \neg\neg Y]$ type : m^2

$X \rightarrow [Y \rightarrow Z]$ type : m^3

$[X \rightarrow Y] \rightarrow Z$ type : m^3

$2m^3 + 6m^2 + m$ well-formed formulae of length 5.

Valuations

Truth table

ϕ	ψ	$\bot$	$\top$	$\neg\phi$	$\phi \to \psi$	$\phi \vee \psi$	$\phi \wedge \psi$	$\phi \mid \psi$	$\cdots$
F	F	F	T	T	T	F	F	T	
	T				T	T	F	T	
T	F			F	F	T	F	T	
	T				T	T	T	F	

T/F assignment for variables determines the valuation of well-formed formulae in a propositional logic system.

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$X \rightarrow [X \rightarrow X]$ type : m

$[X \rightarrow Y] \rightarrow Z$ type

$2m^2 + 3m$ tautologies of length 5.

Density of tautologies

Let $p_{m,n}$ be the probability of tautologies among length n well-formed formulae with m variables.

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Existence of the density

For any fixed number m of variables, the density of tautologies $\tau_m := \lim_{n\to\infty} p_{m,n}$ exists.

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Question

What is the asymptotic behavior of τ_m?

Known results

Fournier, Gardy, Genitrini, and Zaionc, 2010

For the logic system with the implication, m variables and their negative literals, both asymptotic behavior of the density of tautologies and the density of simple tautologies are

$$\frac{7}{8m} + O\left(\frac{1}{m^2}\right)$$

E., *preprint*

For the logic system with the implication and negation, the asymptotic behavior of the density of simple tautologies is

$$\frac{1}{m} + O\left(\frac{1}{m\sqrt{m}}\right)$$

New result

The density of tautologies

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Expected valuation

If we have a well-formed formula which is expected to be true/false, then we can define expected valuations.

$\neg\phi$ is True	ϕ is False
$\neg\phi$ is False	ϕ is True
$\phi \to \psi$ is False	ϕ is True, ψ is False
$\phi \to \psi$ is True	ϕ is False **or** ψ is True

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We can repeat this process until we reach the variables.

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$$[[\neg X \rightarrow \neg Y] \rightarrow [Y \rightarrow X]]{:}\mathrm{F}$$

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$$(\neg X{:}\mathrm{F} \textbf{ or } \neg Y{:}\mathrm{T}),\ Y{:}\mathrm{T},\ X{:}\mathrm{F}$$

Backward valuation

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$$(X{:}\mathrm{T},\ Y{:}\mathrm{T},\ X{:}\mathrm{F}) \textbf{ or } (Y{:}\mathrm{F},\ Y{:}\mathrm{T},\ X{:}\mathrm{F})$$

Shape

A shape is a template for generating well-formed formulae.

$$[\neg X_1 \to \neg X_2] \to [X_3 \to X_4]$$

Here, $X_1, \cdots, X_4$ are placeholder for variables.

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As we observed, this shape is a tautology if $X_1 = X_4$ and $X_2 = X_3$.

L-valuation and R-valuation

For a falsely expected shape, we have two special expected valuations.

- If $[\phi \to \psi]$:T, then always choose ϕ:F.
- If $[\phi \to \psi]$:T, then always choose ψ:T.

We will call the first one as the L-valuation, the later one as the R-valuation, and the common expected valuation as the C-valuation.

Generating functions

Generating function for well-formed formulae

The generating function $W(z)$ for well-formed formulae is defined as

$$W(z) = \sum_{\phi} z^{\text{length}(\phi)}$$

Then,

$$W(z) = mz + zW(z) + zW(z)^2$$

Here,

- mz refers to variables.
- $zW(z)$ refers to $\neg\phi$ form well-formed formulae.
- $zW(z)^2$ refers to $\phi \rightarrow \psi$ form well-formed formulae.

Generating functions

For a fixed shape,

- ν_F be the number of falsely expected placeholders for the C-valuation.
- ν_T be the number of truely expected placeholders for the C-valuation.

Generating function for C-valuations.

The generating function $F(z, s, t)$ is defined as

$$F(z, s, t) = \sum_{\phi} s^{\nu_F} t^{\nu_T} z^{\text{length}(\phi)}$$

Then,

$$F(z, s, t) = mzs + z(1 + F(z, s, t))(mzt + zF(z, s, t) + zW(z)^2).$$

Probability estimation

For a fixed shape, let

- ν_L be the number of placeholders for the L-valuation.
- ν_R be the number of placeholders for the R-valuation.

Probability of tautologies with a fixed shape

For a fixed shape, probability of tautologies with given shape is bounded by

$$\frac{\nu_F \nu_T}{m} + \frac{v_L^2 v_R^2}{16 m^2}$$

Probability estimation

Probability of tautologies

The probability of tautologies of the length n is bounded by

$$\frac{1}{m}\frac{1}{[z^n]W(z)}\frac{\partial}{\partial s}\frac{\partial}{\partial t}[z^n]F(z,s,t)\Big|_{(1,1)} + O(m^{-3/2}).$$

This comes from $\mathbb{E}[\mathbb{E}[A|B]] = \mathbb{E}[A]$.

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Szegő lemma

If W_* is a generating function on a subset of well-formed formula,

$$\lim_{n\to\infty}\frac{[z^n]W_*(z)}{[z^n]W(z)} = \lim_{z\to\frac{1}{1+2\sqrt{m}}-}\frac{W_*'(z)}{W'(z)}.$$

if both exist.

The density of tautologies

Result

The density of tautologies τ_m among well-formed formulae with m variables is

$$\tau_m = \frac{1}{m} + O\left(\frac{1}{m\sqrt{m}}\right)$$

Application

Algorithm

Suppose that we consider the following simple probabilistic algorithm:

- For k random valuations, check whether ϕ is true for all of them.
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With some more detailed computations, there is a constant C such that if $k \geq \frac{C}{\epsilon} m(\log m)^2$, then the algorithm is correct with probability at least $1 - \epsilon$.

Thank you.